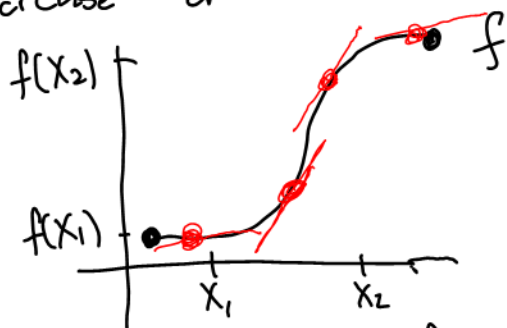


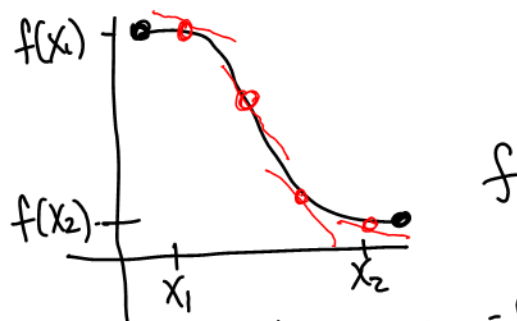
^{The}
Lecture 17: I/D and 1st Derivative Test

Goal: Use the 1st derivative to determine properties of the original function.

Recall what it means for a function f to increase or decrease on an interval.



f is increasing if
whenever $x_1 < x_2$, $f(x_1) < f(x_2)$



f is decreasing if
whenever $x_1 < x_2$, $f(x_1) > f(x_2)$

Theorem (I/D Test) Let f be a differentiable function on an open interval I .

① If $f'(x) > 0$ for every x in I , f is increasing on I .

② If $f'(x) < 0$ for every x in I , f is decreasing on I .

Ex 1 When is the function $f(x) = \frac{x^2}{1+x^2}$ inc and dec?

Step 1 Determine the critical number(s) of f .

$$f'(x) = \frac{2x(1+x^2) - (2x)(x^2)}{(1+x^2)^2} = \frac{2x + 2x^3 - 2x^3}{(1+x^2)^2}$$

$$= \frac{2x}{(1+x^2)^2} \quad \underline{\text{set}} \quad \underline{0}$$

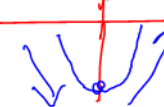
$$2x = 0$$

$$x = 0$$

Step 2 Use the critical number(s) to divide the x -axis into region and determine the sign of the derivative



Test Point:	-1	1
Sign of $2x$:	$2(-1) = -2$ -	+
Sign of $(x^2+1)^2$:	+	+
Sign of f' :	-	+

Results of 1D Test: Dec  Inc

Conclusion: f is increasing on $(0, \infty)$, while f is decreasing on $(-\infty, 0)$

f has a rel. min at $x=0$.

NOTE: Some people just use a number line to organize information



Example 2 Let f be a polynomial s.t.
 $f'(x) = (x-2)(x+1)(x-4)$

Step 1 Determine crit. nums.
 $(x-2)(x+1)(x-4) \stackrel{\text{set}}{=} 0$

$$\Rightarrow x = 2, -1, 4 \Rightarrow x = -1, 2, 4$$

Step 2 Make "sign chart"

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f is increasing on $(-1, 2) \cup (4, \infty)$
 f is decreasing on $(-\infty, -1) \cup (2, 4)$

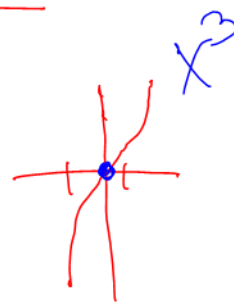
f has a rel. max at 2,
has a rel. min at $x = -1, 4$

Example 3 Repeat for $f(x) = x^3$
 $f'(x) = 3x^2 \stackrel{\text{set}}{=} 0 \Rightarrow x = 0$

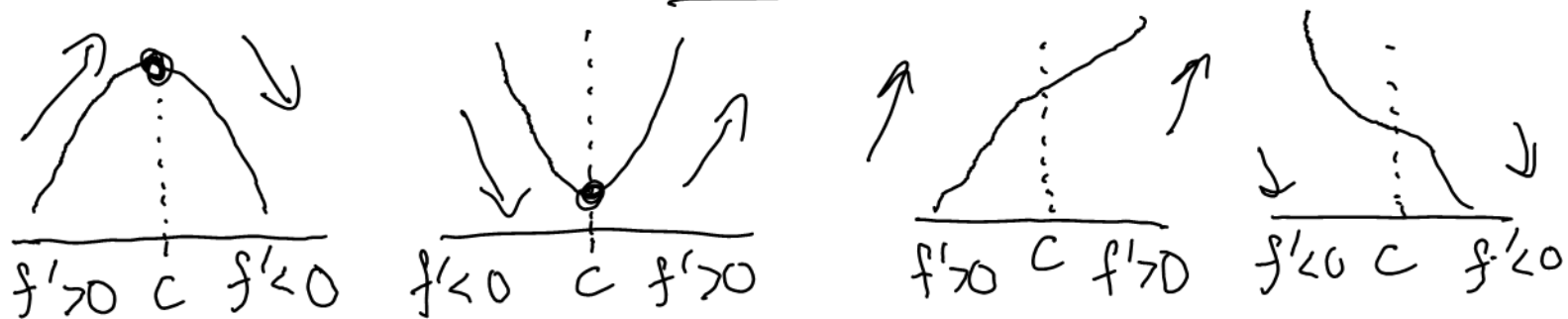
		$\leftarrow$							$\rightarrow$
			-1		0		1		
Test Point									
Sign of f'			+				+		
			Inc	$\nearrow$	$\nearrow$		Inc		

Increasing on $(-\infty, \infty)$ as opposed to
 $(-\infty, 0) \cup (0, \infty)$

x^3 has neither a rel. max/min at 0.



The ID Test gives us a way to determine the location of relative extrema (rel. max and min)



Theorem (1st Derivative Test) Let c be a crit.

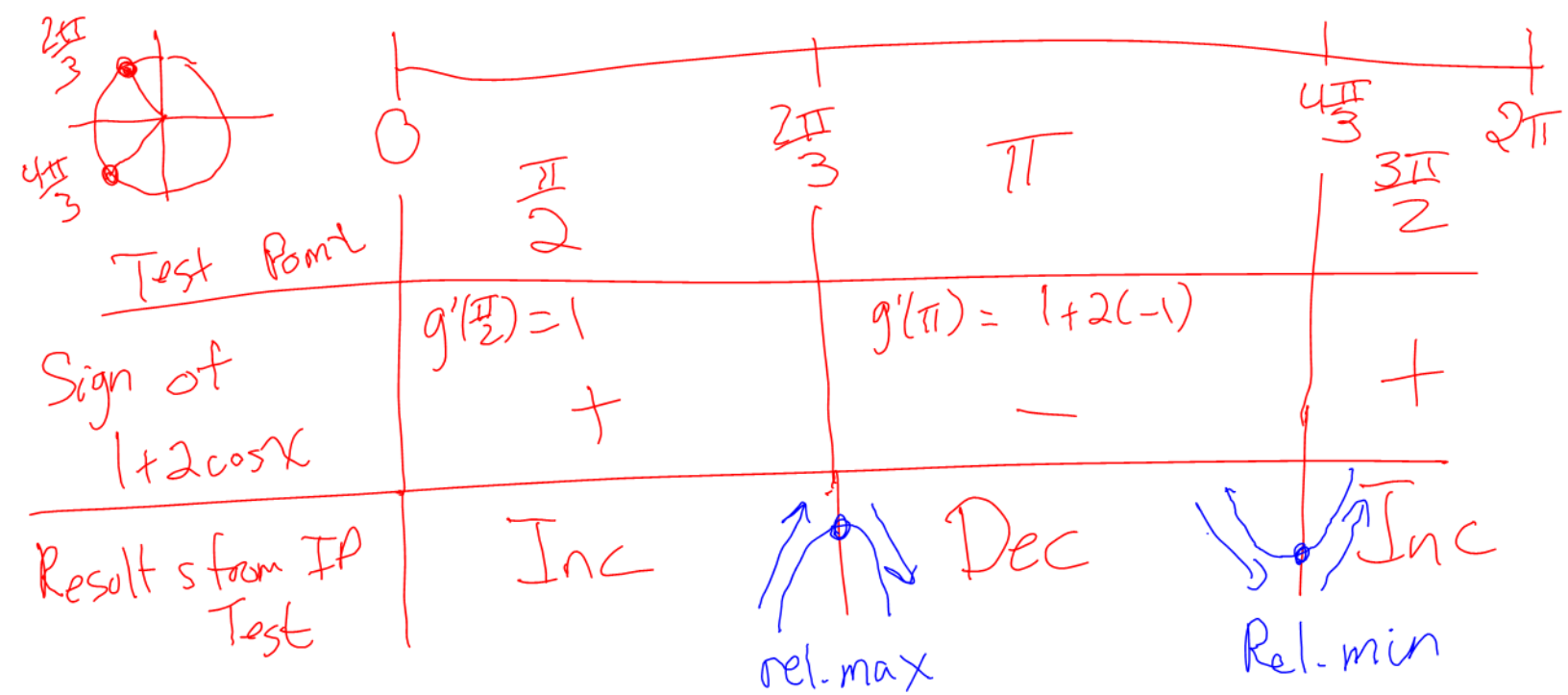
num. of a diff. fun. f

- ① If f' switches from $+$ to $-$ at c , then f has a rel. max. at c .
- ② If f' switches from $-$ to $+$ at c , then f has a rel. min at c .
- ③ If f' does not switch signs at c , f has neither a rel. max nor rel. min at c .

Ex 4 The crit. nums of $g(x) = x + 2\sin x$ on $(0, 2\pi)$ are $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. Determine any rel. ext. on $(0, 2\pi)$

Step 1 $g'(x) = 1 + 2\cos x$

Step 2 Make sign Chart



Ex 5 Repeat for

$$f(x) = \frac{1}{8} e^{2x} (4x^3 - 6x^2 - 6x + 3)$$

$$f'(x) = e^{2x} x (x^2 - 3)$$